12 Mrs Atkins is going to choose two students from her class to take part in a competition.
She can choose from 16 girls and 14 boys.

(a) Work out the number of different ways of choosing one girl and one boy.

$16 \times 14 = 224$ ways

1 mark
One choice from each.

.....

(1)

(b) Work out the number of different ways of choosing two boys.

First boy: 14 choices; second boy: 13.

1 mark
Without replacement.

$14 \times 13 = 182$ ordered choices.

Each pair counted twice, so divide by 2.

$182 \div 2 = 91$ ways

1 mark
Unordered pairs: 91.

.....

(2)

(Total for Question 12 is 3 marks)